Research Statement

Gabriel Angelini-Knoll

I am an algebraic topologist and my work sheds light on the interplay between algebraic topology, number theory, and geometric topology. I study invariants of rings, and more generally $\mathbb{E}_n$-rings, such as algebraic K-theory and topological Hochschild homology, which provide a bridge between these fields. The fields of chromatic homotopy theory and equivariant homotopy theory are also at the heart of my research program.

Algebraic K-theory assigns a graded abelian group $\mathrm{K}_*(R)$ to a ring R. In low degrees, it recovers invariants such as the class group, the Picard group, and the Brauer group. In high degrees, algebraic K-theory groups encode special values of zeta functions. The low degree algebraic K-theory groups also recover classical obstruction groups in geometric topology, such as Wall's finiteness obstruction and the Whitehead group of a manifold. The higher algebraic K-theory groups encode the homotopy groups of the stable h-cobordism space of a manifold. Topological Hochschild homology (THH) is a first approximation to algebraic K-theory. One constructs topological cyclic homology (TC) out of extra structure on THH and TC is a close approximation to algebraic K-theory. The invariant THH also has important connections to number theory through its connection to the de Rahm–Witt complex and integral p-adic Hodge theory. It also has applications to geometric topology, for example it can be used to build certain knot Floer homology theories.

Chromatic homotopy provides a filtration of homotopy groups into periodic families of elements of increasing wavelength, where longer wavelength corresponds to higher chromatic complexity and deeper arithmetic information. In particular, the α-family in the homotopy groups of spheres encodes special values of the Riemann zeta function. The β-family is a periodic family of longer wavelength and it encodes integral modular forms. The Lichtenbaum–Quillen conjectures exhibit deep connections between number theory and algebraic K-theory. Ausoni–Rognes extended these conjectures to $\mathbb{E}_1$-rings using chromatic stable homotopy theory. Higher chromatic complexity corresponds to periodicity of longer wavelength and consequently it is more red. The imprecise version of the Ausoni–Rognes conjectures states that algebraic K-theory increases chromatic complexity by one and therefore they are referred to as the red-shift conjectures.

My research gives evidence for the red-shift conjectures from several different perspectives. I have done this by developing tools for computing topological Hochschild homology [AK18], giving new computations of topological Hochschild homology using these tools [AK17, AK21b, AKCH21b, AKCH21a], and building on these computations to give evidence [AK18, AKS20, AKQ19, AK21a, AKAC$^+$21, AKHW21] for the red-shift conjectures in algebraic K-theory. These results are summarized in Section 1.

I also work in the area of equivariant homotopy theory with applications to geometric topology. I prove the Segal conjecture for topological Hochschild homology for a certain family of $\mathbb{E}_1$-rings [AKQ21] with applications to the stable h-cobordism space of a point. My work gives new perspectives on equivariant topological Hochschild homology and equivariant Witt vectors [AKGH21, AKMP21] with applications to equivariant knot Floer homology. I also use synthetic and motivic homotopy theory to study C_3-equivariant homotopy, with the ultimate goal of resolving the 3-primary Arf–Kervaire invariant problem [AKBBK21]. These results are summarized in Section 2.

1 Arithmetic of $\mathbb{E}_n$-rings

This section is split into four subsections: Section 1.1 a brief overview of the red-shift conjectures, Section 1.2 a brief overview of trace methods, Section 1.3 a summary of results, and Section 1.4 a summary of work in progress.

1.1 Higher Lichtenbaum–Quillen conjectures

Broadly, the Ausoni–Rognes red-shift conjectures are about the arithmetic of $\mathbb{E}_n$-rings, which generalize rings. To discuss these conjectures, I must fix some terminology from chromatic homotopy theory. Write $K(n)$ for the Morava K-theory spectra at a fixed prime p. These $\mathbb{E}_1$-rings are the prime fields in the category spectra. Given a finite spectrum V, we say V is type n if $K(n-1)_*V = 0$, but $K(n)_*V \neq 0$. By [HS98], such a spectrum exists for each n and any such spectrum is equipped with a v_n-power self map $v\colon \Sigma^d V \to V$. We write $T(n)$ for the telescope of this v_n-power self map. The Ausoni–Rognes red-shift conjectures generalize the Lichtenbaum–Quillen conjectures to higher chromatic heights in the following sense. In the phrasing of Waldhausen [Wal84], the Lichtenbaum–Qullen conjectures state that the map

$$\pi_s \mathrm{K}(R[1/p];\mathbb{Z}_p) \longrightarrow \pi_s L_1^f \mathrm{K}(R[1/p];\mathbb{Z}_p)$$

is an isomorphism for $s \gg 0$, where $\mathrm{K}(-;\mathbb{Z}_p)$ denotes the p-complete algebraic K-theory spectrum and L_n^f denotes Bousfield localization at $T(0) \oplus \cdots \oplus T(n)$. This is interesting because Thomason [Tho85] showed that $\pi_* L_1^f \mathrm{K}(-;\mathbb{Z}_p)$ satisfies étale descent and Wiles [Wil90] has demonstrated that quotients of orders of étate cohomology groups of $\mathcal{O}_F$ recover special values of the Dedekind zeta function ζ_F, where F is a number field with ring of integers $\mathcal{O}_F$. One phrasing of the Ausoni–Rognes red-shift conjectures is the following question.

Question 1.1. Let R be an $\mathbb{E}_1$-ring. If $\pi_s R_p \longrightarrow \pi_s L_n^f R_p$ is an isomorphism for $s \gg 0$, is the map

$$\pi_s \mathrm{K}(R;\mathbb{Z}_p) \longrightarrow \pi_s L_{n+1}^f \mathrm{K}(R;\mathbb{Z}_p)$$

an isomorphism for $s \gg 0$?

This reduces to the Lichtenbaum–Quillen conjecture when $n = 0$. A weaker form of this conjecture has been posed for $\mathbb{E}_2$-rings.

Question 1.2. IF R is an $\mathbb{E}_2$-ring, then is

$$K(n)_*R = 0 \iff K(n+1)_*K(R) = 0?$$

One of the original phrasings of the Ausoni–Rognes red-shift conjectures appeared in [Rog00]. This used the notion of FP-type due to [MR99]. We say X has *FP-type* n if

$$n = \min\{\mathrm{type}(F) - 1 : \pi_* F \wedge X \text{ is finite}\}.$$

where F is a finite spectrum. Moreover, we say X has *pure FP-type* n, a stronger condition, if there is a finite type n spectrum V such that V_*X is a finitely generated free $P(v_n)$-module. Rognes' original red-shift question was the following:

Question 1.3. If R has (pure) FP-type n, then does $K(R)$ has (pure) FP-type $n+1$?

1.2 Trace methods

My approach to algebraic K-theory uses the theory of trace methods developed by [BHM93], where one approximates algebraic K-theory by topological cyclic homology. Topological cyclic homology is built using the cyclotomic structure on topological Hochschild homology. Topological Hochschild homology of an $\mathbb{E}_1$-ring R is the factorization homology

$$\mathrm{THH}(R) := \int_{\mathbb{T}} R$$

for the circle $\mathbb{T}$. The topological group $\mathbb{T}$ acts on this by functoriality. Moreover, there is a $\mathbb{T}$-equivariant Tate-valued Frobenius homomorphism

$$\varphi_p \colon \ \mathrm{THH}(R) \longrightarrow \mathrm{THH}(R)^{t\mu_p} \tag{1}$$

where $\mu_p \subset \mathbb{T} \subset \mathbb{C}$ is the subgroup of p-th roots of unity. We write

$$\mathrm{TC}^-(R;\mathbb{Z}_p) := \left(\mathrm{THH}(R)^{h\mathbb{T}}\right) \ \text{ and } \ \mathrm{TP}(R;\mathbb{Z}_p) := \left(\mathrm{THH}(R)^{t\mathbb{T}}\right)^\wedge_p$$

for the p-completions of *topological negative cyclic homology* and *topological periodic cyclic homology* respectively. Then by [NS18], topological cyclic homology $\mathrm{TC}(R;\mathbb{Z}_p)$ is the equalizer

$$\mathrm{TC}^-(R;\mathbb{Z}_p) \underset{\varphi_p^{h\mathbb{T}}}{\overset{can}{\rightrightarrows}} \mathrm{TP}(R;\mathbb{Z}_p)$$

when R is a bounded below $\mathbb{E}_1$-ring. Alternatively, there are restriction maps

$$R \colon \ \mathrm{THH}(R)^{\mu_{p^n}} \longrightarrow \mathrm{THH}(R)^{\mu_{p^{n-1}}}$$

and we define

$$\mathrm{TR}(R) := \lim_{R,n} \mathrm{THH}(R)^{\mu_{p^n}}.$$

Then $\mathrm{TC}(R;\mathbb{Z}_p)$ may also be described as the equalizer

$$\mathrm{TR}(R) \underset{F}{\overset{\mathrm{id}}{\rightrightarrows}} \mathrm{TR}(R)$$

where F is induced by the inclusion of fixed points. Then [DGM13] proved that topological cyclic homology is a close approximation to algebraic K-theory in the sense that if $f \colon \ A \to B$ is a map of $\mathbb{E}_1$-rings such that $\pi_0 f$ is surjective with nilpotent kernel, then writing $E(A,B)$ for the fiber $E(A) \to E(B)$ for $E \in \{K, \mathrm{TC}\}$ we have an equivalence

$$\mathrm{K}(A,B) \simeq \mathrm{TC}(A,B).$$

Besides computing algebraic K-theory, the invariants THH, TC, TC$^-$, TP, and TR have independent arithmetic interest by [HM03, BMS19], for example.

1.3 Results on higher Lichtenbaum–Quillen conjectures

One of my main motivations has been to give evidence for the Ausoni–Rognes red-shift conjectures with the goal of understanding the arithmetic of $\mathbb{E}_n$-rings, which generalize rings. There are potential v_n-periodic families of elements in the p-local stable homotopy groups of spheres known as the p-primary n-th Greek letter family, $\alpha^{(n)}$. When $p \geq 3$ and $n = 1$, the α-family has a close relationship to special values of the Riemann zeta function by work of Adams [Ada66] and this family of elements is detected in the algebraic K-theory of the integers by Quillen [Qui72]. These results were one of the results that motivated Lichtenbaum [Lic73] to speculate that algebraic K-theory groups and special values of zeta functions should be closely related more generally. In [Lau99, Beh06, BL09], Behrens and Laures demonstrated a tight connection between the p-primary β-family for $p \geq 5$ and integral modular forms. The following result is therefore a higher chromatic height analogue of the results of Adams and Quillen.

Theorem 1.4 ([AKS18])**.** The p-primary β-family modulo (p, v_1) is detected in the iterated algebraic K-theory $\pi_* \mathrm{K}(\mathrm{K}(\mathbb{Z}))$ of the integers for $p \geq 5$.

This result built on computations in my PhD thesis [AK17], published in [AK21b] and joint work with Salch [AKS18]. In [AKS18], we provided a useful tool for computing factorization homology of $\mathbb{E}_\infty$-rings. Let $\mathbb{N}$ denote natural numbers with the usual linear ordering and let Sp denote the category of symmetric spectra.

Theorem 1.5 ([AKS18])**.** Let $I \in \mathrm{CAlg}(\mathrm{Fun}(\mathbb{N}, \mathsf{Sp})$, then there is a spectral sequence

$$\mathcal{H}_* \int_M E_0^* I \implies \mathcal{H}_* \int_M I_0$$

for any generalized homology theory $\mathcal{H}$ and framed manifold M, where $E_0^* I$ is the associated graded of the filtered spectrum I.

In [AK21b], I used this to compute mod (p, v_1) topological Hochschild homology of the connective cover of the $K(1)$-local sphere, denoted simply j, at primes $p \geq 5$.

Theorem 1.6 ([AK21b])**.** Let $p \geq 5$. There is an isomorphism of graded $\mathbb{F}_p$-algebras

$$V(1)_* \mathrm{THH}(j) \cong H(E(\alpha_1, \lambda_1, \lambda_2); d) \otimes P(\mu_2) \otimes \Gamma(\sigma b).$$

Towards answering question (1.2), I proved several $K(n)$-local vanishing results. Let $y(n)$ be the Thom spectrum of a map

$$\Omega J_{2^n}(S^2) \longrightarrow \Omega^2 J(S^2) \longrightarrow BO \longrightarrow BGL_1\mathbb{S}$$

Theorem 1.7 ([AKQ19])**.** Whenever the map $y(n) \to H\mathbb{F}_p$ is a $K(m)$-local equivalence, $K(m)_* \mathrm{K}(y(n), \mathbb{F}_p) = 0$. Additionally, $K(m)_* y(n) = 0$ implies that $K(m+1)_* \mathrm{TP}(y(n)) = 0$.

This result uses a technical tool proven in joint work with Salch.

Theorem 1.8 ([AKS20])**.** Given a sequence of spectra Y_i satisfying:

1. the mod p homology of Y_i and $\lim_i Y_i$ have bounded above comodule primitives,
2. each Y_i is p-complete and $\pi_* Y_i$ is finite type over $\mathbb{Z}_p$,
3. the groups $\lim_i H(H_*(Y_i;\mathbb{F}_p);Q_n)$ vanish,

then $L_{K(n)} \lim_i Y_i = 0$.

Using this result and trace methods, we also prove an upper Morava K-theory vanishing region for topological modular forms with level structure $\mathrm{tmf}_1(3)$ and a version of topological automorphic forms taf^D, which are $\mathbb{E}_\infty$-rings that are forms of $BP\langle 2\rangle$ by [LN12, HL10].

Theorem 1.9 ([AKS20])**.** There are equalities $K(m) \oplus K(R) = 0$ for $R \in \{\mathrm{tmf}_1(3), \mathrm{taf}^D\}$ and $m \geq 4$.

The calculations of $\mathrm{THH}_*(BP\langle n\rangle)$ for $n \geq -1$ are deep results and they are increasingly difficult as n increases. For example, Bökstedt's result [Bök87] $\mathrm{THH}_*(BP\langle -1\rangle) = P(\mu_0)$ can be used to prove Bott periodicity [LH19] and the calculation of $\mathrm{THH}_*(BP\langle 1\rangle)$ complicated torsion which "follows a kind of tower-of-Hanoi pattern with increasingly complicated, inductively built, components" [AHL10, p.6]. In joint work with Culver and Höning, we aim to give a complete description of $\mathrm{THH}_*(\mathrm{taf}^D)$ where taf^D is an E_∞ form of $BP\langle 2\rangle$ at $p = 3$ by [HL10]. Towards this aim, we have already proven.

Theorem 1.10 ([AKCH21b])**.** When $m = 0, 1, 2$, there are isomorphisms

$$\mathrm{THH}_*(\mathrm{taf}^D; k(m)) \cong P(v_m) \otimes E(\lambda_1, \ldots, \lambda_{2-m}) \oplus T_m^n,$$

where T_m^n is an explicit v_m-torsion module.

In work in progress, we combine the Hochschild–May spectral sequence

$$\mathrm{THH}_*(H\pi_* \mathrm{taf}^D; H\pi_*\ell) \implies \mathrm{THH}_*(\mathrm{taf}^D, \ell)$$

with the Bockstein spectral sequences

$$\mathrm{THH}_*(\mathrm{taf}^D; k(m))[v_m] \implies \mathrm{THH}_*(\mathrm{taf}^D; \ell)$$

to compute $\mathrm{THH}_*(\mathrm{taf}^D; \ell)$ in [AKCH21a].

1.4 Work in progress on higher Lichtenbaum–Quillen conjectures

Currently, I am working on two projects that extend the current knowledge of the red-shift conjectures. In joint work with Hahn and Wilson, we study the red-shift conjecture for Morava K-theory.

Theorem 1.11 ([AKHW21])**.** Question 1.1 is true for $R = K(n)$ at all primes and all for all $n \geq 0$. Also, Question 1.2 is false for $K(n)$.

In joint work with Ausoni, Culver, Höning, and Rognes, we study mod (p, v_1, v_2)-homotopy of $\mathrm{K}(BP\langle 2\rangle)$.

Theorem 1.12 ([AKAC+21])**.** Algebraic K-theory of any $\mathbb{E}_3$-ring form of $BP\langle 2\rangle$ has pure FP-type 3 at $p \geq 7$, in other words the strong form of Question 1.3 holds for $BP\langle 2\rangle$.

This completes work begun by Ausoni and Rognes in the early 2000's and uses recent work of Hahn–Wilson [HW20] proving that $BP\langle 2\rangle$ is an $\mathbb{E}_3$-ring.

2 Geometry of $\mathbb{E}_n$-rings

This section is split into four sub-sections: Section 2.1 a brief survey of terms in equivariant topology, Section 2.2 on the Segal conjecture for topological Hochschild homology, Section 2.3 on combinatorial equivariant topological Hochschild homology, and Section 2.4 on work in progress in motivic homotopy theory.

2.1 Equivariant algebra and topology

Fix a finite group G and a G-spectrum X. There are structure maps called the transfer maps $\mathrm{tr}_K^H\colon X^K \to X^H$, restriction maps $\mathrm{res}_K^H\colon X^H \to X^K$ and a Weyl group action $W_G(H)$ on X^H. In fact, one can completely characterize a G-spectrum in terms of this data and compatibility of these maps [GM11, Bar14]. The algebraic analogue of this structure is a Mackey functor, which essentially consists of an abelian group $\underline{M}(G/H)$ for each G-set G/H together with restriction and transfer maps and Weyl group actions satisfying the same compatibility. The category of G-Mackey functors Mak_G is closed symmetric monoidal with monoidal product called the box product.

When considering algebras, the theory is more subtle than in the non-equivariant setting. For example, there are (at least) two notions of commutative ring objects: commutative monoids with respect to box product known as commutative Green functors and Tambara functors, which have an additional structure map $\mathrm{Norm}_H^G\colon \underline{M}(G/H) \to \underline{M}(G/G)$. Restricting attention to the cyclic group of order 2, denoted D_2, there are two notions of associative algebra as well. There are associative monoids with respect to the box product and there are discrete $\mathbb{E}_\sigma$-rings, which we will define shortly. We first note that there is a Norm functor on G-spectra given by the composite

$$N_H^G\colon \mathsf{Sp}^H \xrightarrow{\wedge^{|G/H|}} \mathsf{Sp}^{H\wr\Sigma_n} \xrightarrow{\alpha^*} \mathsf{Sp}^G$$

where α^* is induced by the group homomorphism $\alpha\colon G \longrightarrow H\wr\Sigma_n$ associated to a choice of ordering on the space of cosets G/H. When restricted to commutative monoids the functor

$$N_H^G\colon \ \mathrm{CAlg}(\mathsf{Sp}^H) \longrightarrow \mathrm{CAlg}(\mathsf{Sp}^G)$$

is left adjoint to the restriction functor ι_e^*.

By an $\mathbb{E}_\sigma$-ring in Sp^{D_2}, we mean an algebra R over the monad

$$\mathcal{P}_\sigma(R) := T(N_e^{D_2}\iota_e^* R) \otimes (\mathbb{S} \oplus R)$$

and by a discrete $\mathbb{E}_\sigma$-ring we mean an algebra over the monad $\underline{\pi}_0^{D_2}\mathcal{P}_\sigma H-$ where H is the Eilenberg–MacLane functor. The main examples of $\mathbb{E}_\sigma$-rings are $\mathbb{E}_1$-rings R with anti-involution $\tau\colon R^{\mathrm{op}} \to R$. They are important for us because they are the input of Real topological Hochschild homology, a equivariant analogue of topological Hochschild homology. More generally, given a group homomorphism $G_0 \to D_2$ we can define an $\mathbb{E}_{\widetilde{\sigma}}$-ring in G_0-spectra where $\widetilde{\sigma}$ is the inflation of σ along the group homomorphism $G_0 \to D_2$.

2.2 The Segal conjecture

At the heart of both the arithmetic of $\mathbb{E}_n$-rings and the geometry of $\mathbb{E}_n$-rings is the algebraic K-theory of the sphere spectrum $\mathrm{K}(\mathbb{S})$. First, algebraic K-theory of any $\mathbb{E}_1$-ring receives a $\mathrm{K}(\mathbb{S})$-module map $\mathrm{K}(\mathbb{S}) \to \mathrm{K}(R)$. On homotopy groups, the map $\mathrm{K}_s(\mathbb{S}) \to \mathrm{K}_s(\mathbb{Z})$ splits so all of the arithmetic encoded in algebraic K-theory of the integers also appears in algebraic K-theory of $\mathbb{S}$. Moreover, Algberaic K-theory of the sphere spectrum splits as

$$\mathrm{K}(\mathbb{S}) = \mathbb{S} \oplus \mathrm{Wh}^{\mathrm{DIFF}}(*) \tag{2}$$

where $\mathrm{Wh}^{\mathrm{DIFF}}(*)$ is a delooping of stable h-cobordism space $\mathcal{H}(*)$. The stable homotopy groups of spheres are also a summand of the algebraic K-theory of the the sphere as well by (2), which are the cornerstone of stable homotopy theory.

In [DR17], Dundas–Rognes suggest a program for computing algebraic K-theory of A in terms of algebraic K-theory of B by descent along the map $A \to B$ in certain cases. Let $X(n)$ is the Thom spectrum of a 2-fold loop map $\Omega SU(n) \longrightarrow BGL_1\mathbb{S}$ and therefore an $\mathbb{E}_2$-ring with a map of $\mathbb{E}_2$-rings $\mathbb{S} \to X(n)$. Consider the Amitsur complex

$$\mathbb{S} \longrightarrow X(n) \rightrightarrows X(n)^{\otimes 2} \substack{\longrightarrow \\ \longrightarrow \\ \longrightarrow} \dots$$

then Dundas–Rognes [DR17] prove that $\mathrm{K}(\mathbb{S}) = \mathrm{Tot}\left([n] \mapsto K(X(n)^{\otimes n+1})\right)$ so one may use $\mathrm{K}_*(X(n))$ to approximate $\mathrm{K}_*(\mathbb{S})$.

In most computations of algebraic K-theory, one of the key steps in the proof is the Segal conjecture, which we phrase as the question of whether the Tate valued Frobenius map (1) has co-connective fiber after p-completion. In joint work with Quigley [AKQ19], I prove a strong form of the Segal conjecture for THH of the family of spectra $X(n)$.

Theorem 2.1 ([AKQ19])**.** The map

$$\varphi_p\colon\ \mathrm{THH}(X(n)) \longrightarrow \mathrm{THH}(X(n))^{tC_p}$$

is an equivalence after p-completion for all primes p. Consequently, the map

$$\varphi_p^{h\mathbb{T}}\colon\ \mathrm{TC}^-(X(n);\mathbb{Z}_p) \to \mathrm{TP}(X(n);\mathbb{Z}_p)$$

is an equivalence for all primes p.

2.3 Combinatorial equivariant factorization homology

In [ABG+18], they define the norm $N_e^{\mathbb{T}}(R)$ for the circle group $\mathbb{T}$ as the cyclic bar construction of R and prove that it satisfies the universal property of the norm. They also define norms $N_{\mu_n}^{\mathbb{T}}(R)$ where μ_n is the n-th roots of unity in $\mathbb{T} \subset \mathbb{C}$. In [BGHL18], they use this theory to define a new theory of Witt vectors and Hochschild homology for associative μ_n-Green functors.

As discussed in Section 2.1, there is more than one notion of ring in D_2-spectra Sp^{D_2}, where D_2 is the cyclic group of order 2: namely $\mathbb{E}_1$-rings and $\mathbb{E}_\sigma$-rings. Given an $\mathbb{E}_\sigma$-ring R, we define the norm $N_{D_2}^{O(2)}(R)$ to be the dihedral bar construction $B^{\mathrm{di}}(R)$ of R. In [AKGH21], we give the following evidence that this is a reasonable definition.

Theorem 2.2 ([AKGH21])**.** The restriction of the functor $N_{D_2}^{O(2)}$ to $\mathrm{CAlg}(\mathsf{Sp}^{D_2})$

$$N_{D_2}^{O(2)}\colon\ \mathrm{CAlg}(\mathsf{Sp}^{D_2}) \longrightarrow \mathrm{CAlg}(\mathsf{Sp}^{O(2)})$$

is left adjoint to the restriction functor ι_e^*.

Another property of the norm for finite groups is the multiplicative double coset formula. For our new notion of norm $N_{D_2}^{O(2)}$, we also prove a multiplicative double coset formula, extending work of [DMPP17].

Theorem 2.3 ([AKGH21])**.** Under cofibrancy hypotheses on R, there is an equivalence of D_{2m}-spectra

$$\iota_{D_{2m}}^* N_{D_2}^{O(2)}(R) \simeq N_{D_2}^{D_{2m}} R \otimes_{N_e^{D_{2m}} \iota_e^* R} N_{\triangle}^{D_{2m}} \iota_{\triangle}^* c_\zeta R$$

where ζ is the generator of $\mu_{2m} \subset D_{4m}$ and $\triangle = \zeta D_2 \zeta^{-1}$.

We use these results to motivate a new definition of Real Hochschild homology $\underline{\mathrm{HR}}_*^{D_{2m}}(\underline{M})$ of a discrete $\mathbb{E}_\sigma$-ring $\underline{M}$, or ring with anti-involution. We prove that this new theory is equipped with a type of cyclotomic structure given by structure maps

$$R\colon\ \underline{\mathrm{HR}}_0^{D_{2p^k}}(\underline{M})^{\mu_{p^k}} \longrightarrow \underline{\mathrm{HR}}_0^{D_{2p^{k-1}}}(\underline{M})^{\mu_{p^{k-1}}}$$

which are D_2-Mackey functor maps. Here we write $\underline{N}^{\mu_{p^k}}$ for the D_2-Mackey functor $\underline{\pi}_0^{D_2}(H\underline{N})^{\mu_{p^k}}$. There are also Frobenius and Verschiebung maps F and V coming from the equivariant structure and therefore we can define

$$\mathbb{W}(\underline{M}; p) = \lim_R \underline{\mathrm{HR}}_0^{D_{2p^k}}(\underline{M})^{\mu_{p^k}}$$

and prove the following refinement of a result of Hesselholt [Hes97], where TCR is an equivariant analogue of topological cyclic homology.

Theorem 2.4 ([AKGH21])**.** Given an $\mathbb{E}_\sigma$-ring R with $\underline{\pi}_0^{D_2} R = \underline{M}$ such that a certain $R^1 \lim$ vanishes, then there is an isomorphism

$$\mathrm{TCR}_{-1}(R) = \mathbb{W}(\underline{M}; p)_F$$

where $\mathbb{W}(\underline{M}; p)_F$ is the coinvariants of an operator F on $\mathbb{W}(\underline{M}; p)$.

In joint work with Maximilien Peroux and Mona Merling [AKMP21], we extend the theory of combinatorial equivariant factorization homology to more general compact Lie groups. In particular, a crossed simplicial group (csg), denoted $\Delta\mathfrak{G}$, is a category with the same objects as the simplex category Δ, but with additional automorphisms $\mathfrak{G}_n$ of $[n]$ added in satisfying the condition: every morphism in $\Delta\mathfrak{G}$ can be uniquely written as $g \circ \phi$ where $g \in \mathfrak{G}_n$ and $\phi \in \Delta([m],[n])$. Any simplicial group $G_\bullet$ gives rise to a crossed simplicial group $\Delta\mathfrak{G}$ with $\mathrm{Aut}_{\Delta\mathfrak{G}}([n]) = G_n$. Connes' cyclic category Λ is an example of a crossed simplicial group that does not come from a simplicial group. For any crossed simplicial group, the groups $\mathfrak{G}_n := \mathrm{Aut}_{\Delta\mathfrak{G}}([n])$ form a simplicial set, even though the structure maps are not necessarily group homomorphisms, and the geometric realization $\mathfrak{G} = |\mathfrak{G}_\bullet|$ is a compact

topological group. We have several constructions (a Bökstedt model, a norm model, and an ∞-categorical construction) of topological csg homology of an $\mathbb{E}_{\widetilde{\sigma}}$-ring in $\mathfrak{G}_0$-spectra

$$N^{\mathfrak{G}}_{\mathfrak{G}_0}(R) = \mathrm{TH}\,\mathfrak{G}(R).$$

Of particular interest is the case of the quaternionic crossed simplicial group. In this case, let R be an $E_{\widetilde{\sigma}}$-ring in Q_4-spectra, where Q_4 is the cyclic group of order 4 and $\widetilde{\sigma}$ is the inflation of the representation σ along the canonical quotient $Q_4 \to D_2$. This gives a combinatorial model for the norm

$$N^{\mathrm{Pin}-(2)}_{Q_4}(R) = \mathrm{THQ}(R),$$

which we aim to use to better understand involutive knot Floer homology in analogy with the relationship between topological Hochschild homology and knot Floer homology.

Goal 2.5. Construct a Hodge-to-de Rahm spectral sequence for THQ that can be used to study knot Floer homology.

As a completely distinct question, this new theory has potential to receive a natural trace map from equivariant algebraic K-theory.

Goal 2.6. Construct a trace map

$$\mathrm{K}_G(R) \longrightarrow N^{G\times\mathbb{T}}_G(R).$$

To give a further refinement of the trace map, we aim to extend the theory of cyclotomic spectra to this level of generality.

Goal 2.7. Define $\mathfrak{G}$-cyclotomic structure so that there is a notion of topological $\mathfrak{G}$-cyclic homology

$$\mathrm{TC}\,\mathfrak{G}(R).$$

When $\Delta\mathfrak{G} = \Lambda \times \{G\}$ is the trivial extension of Connes' cyclic category Λ by the crossed simplicial group associated to the constant simplicial group G, we call this TC_G. This theory should refine the cyclotomic trace

$$K_G(R) \longrightarrow \mathrm{TC}_G(R) \to N^{G\times\mathbb{T}}_G(R)$$

and when $\Delta\mathfrak{G}$ is the dihedral category this will recover TCR.

2.4 Synthetic C_p-equivariant homotopy

One can use motivic homotopy theory $\mathrm{SH}(\mathbb{R})$ over $\mathrm{spec}(\mathbb{R})$ to simplify computations in Sp^{D_2}. In the case of a cyclic group of order p, denoted C_p, where p is odd there is not a known good replacement for $\mathrm{SH}(\mathbb{R})$. In work in progress with Behrens, Belmont, and Kong [AKBBK21], I aim to give a purely synthetic construction equipped with a realization functor to Sp^{C_p}. Synthetic homotopy theory was initiated by [GIKR18] and extended significantly by [Pst18]. Recently, this was used by Burkland–Hahn–Senger [BHS20] to extend the cellular $\mathbb{R}$-motivic stable homotopy category in a way that allows for realization functors recovering $RO(C_2)$-graded homotopy groups of C_2-spectra.

Goal 2.8. Combine the theory of recollement in stable homotopy theory and synthetic spectra to give an artificial anologue of $\mathrm{SH}(\mathbb{R})$ equipped with a realization functor to Sp^{C_p} for odd primes p.

Ravenel has suggested that a nice C_p-equivariant Brown–Peterson spectrum BP could be used to resolve the remaining cases of the odd primary Arf–Kervaire invariant one problem in analogy with the solution of the 2-primary Kervaire invariant one problem by [HHR16]. The key part of this being the existence of a nice Adams–Novikov spectral sequence based at this C_p-equivariant BP. Some progress in this direction appears in work of Hahn–Senger–Wilson [HSW20]. In our work, we skip the construction of BP and instead aim to construct an analogous spectral sequence.

Goal 2.9. Resolve the 3-primary Arf–Kervaire invariant one problem using our new artificial 3-primary analogue of $\mathrm{SH}(\mathbb{R})$.

References

[ABG+18] Vigleik Angeltveit, Andrew J. Blumberg, Teena Gerhardt, Michael A. Hill, Tyler Lawson, and Michael A. Mandell. Topological cyclic homology via the norm. *Doc. Math.*, 23:2101–2163, 2018.

[Ada66] J. F. Adams. On the groups $J(X)$. IV. *Topology*, 5:21–71, 1966.

[AHL10] Vigleik Angeltveit, Michael A. Hill, and Tyler Lawson. Topological Hochschild homology of ℓ and *ko*. *Amer. J. Math.*, 132(2):297–330, 2010.

[AK17] Gabriel J. Angelini-Knoll. *Periodicity in Iterated Algebraic K-Theory of Finite Fields*. ProQuest LLC, Ann Arbor, MI, 2017. Thesis (Ph.D.)–Wayne State University.

[AK18] Gabriel Angelini-Knoll. Detecting β elements in iterated algebraic K-theory of finite fields. *Submitted.*, 2018. arXiv:1810.10088.

[AK21a] Gabriel Angelini-Knoll. Complex orientations and TP of complete DVRS. *arXiv e-prints*, page arXiv:2104.07306, April 2021.

[AK21b] Gabriel Angelini-Knoll. On topological Hochschild homology of the $K(1)$-local sphere. *Journal of Topology*, 14(1):258–290, 2021.

[AKAC+21] Gabriel Angelini-Knoll, Christain Ausoni, Dominic Leon Culver, Eva Höning, and John Rognes. Algebraic K-theory of elliptic cohomology. *In progress*, 2021.

[AKBBK21] Gabriel Angelini-Knoll, Mark Behrens, Eva Belmont, and Hana Kong. Synthetic C_p-equivariant homotopy theory. *In progress*, 2021.

[AKCH21a] Gabriel Angelini-Knoll, Dominic Culver, and Eva Höning. Topologial Hochschild homology of truncated Brown-Peterson spectra II. *In progress*, 2021.

[AKCH21b] Gabriel Angelini-Knoll, Dominic Leon Culver, and Eva Höning. Topological Hochschild homology of truncated Brown-Peterson spectra I. *arXiv e-prints*, page arXiv:2106.06785, June 2021.

[AKGH21] Gabriel Angelini-Knoll, Teena Gerhardt, and Michael Hill. Real topological Hochschild homology, Witt vectors, and norms. *Preprint*, 2021.

[AKHW21] Gabriel Angelini-Knoll, Jeremy Hahn, and Dylan Wilson. Red-shift for Morava K-theory. *In progress*, 2021.

[AKMP21] Gabriel Angelini-Knoll, Mona Merling, and Maximilien Péroux. Topological csg homology. *Preprint*, 2021.

[AKQ19] Gabriel Angelini-Knoll and J. D. Quigley. Chromatic complexity of the algebraic K-theory of $y(n)$. *Submitted.*, 2019. arXiv:1908.09164.

[AKQ21] Gabriel Angelini-Knoll and J. D. Quigley. The Segal conjecture for topological Hochschild homology of Ravenel spectra. *Journal of Homotopy and Related Structures*, 2021.

[AKS18] Gabe Angelini-Knoll and Andrew Salch. A May-type spectral sequence for higher topological Hochschild homology. *Algebr. Geom. Topol.*, 18(5):2593–2660, 2018. agt.2018.18.2593.

[AKS20] Gabriel Angelini-Knoll and Andrew Salch. Commuting unbounded homotopy limits with Morava K-theory. *Submitted.*, 2020. arXiv:2003.03510.

[Bar14] C. Barwick. Spectral mackey functors and equivariant algebraic k-theory (i). 04 2014.

[Beh06] Mark Behrens. A modular description of the $K(2)$-local sphere at the prime 3. *Topology*, 45(2):343–402, 2006.

[BGHL18] A. J. Blumberg, T. Gerhardt, M. A. Hill, and T. Lawson. The Witt vectors for Green functors. *ArXiv e-prints*, March 2018.

[BHM93] M. Bökstedt, W. C. Hsiang, and I. Madsen. The cyclotomic trace and algebraic K-theory of spaces. *Invent. Math.*, 111(3):465–539, 1993.

[BHS20] Robert Burklund, Jeremy Hahn, and Andrew Senger. Galois reconstruction of Artin-Tate $\mathbb{R}$-motivic spectra. *arXiv e-prints*, page arXiv:2010.10325, October 2020.

[BL09] Mark Behrens and Gerd Laures. β-family congruences and the f-invariant. In *New topological contexts for Galois theory and algebraic geometry (BIRS 2008)*, volume 16 of *Geom. Topol. Monogr.*, pages 9–29. Geom. Topol. Publ., Coventry, 2009.

[BMS19] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. Topological Hochschild homology and integral p-adic Hodge theory. *Publ. Math. Inst. Hautes Études Sci.*, 129:199–310, 2019.

[Bök87] M. Bökstedt. The topological Hochschild homology of $\mathbb{Z}$ and of $\mathbb{Z}/p\mathbb{Z}$. preprint, 1987.

[DGM13] Bjørn Ian Dundas, Thomas G. Goodwillie, and Randy McCarthy. *The local structure of algebraic K-theory*, volume 18 of *Algebra and Applications*. Springer-Verlag London, Ltd., London, 2013.

[DMPP17] E. Dotto, K. Moi, I. Patchkoria, and S. Precht Reeh. Real topological Hochschild homology. *ArXiv e-prints*, November 2017.

[DR17] B. I. Dundas and J. Rognes. Cubical and cosimplicial descent. *ArXiv e-prints*, April 2017.

[GIKR18] Bogdan Gheorghe, Daniel C. Isaksen, Achim Krause, and Nicolas Ricka. C-motivic modular forms. 10 2018.

[GM11] Bertrand Guillou and J. P. May. Models of G-spectra as presheaves of spectra. *arXiv e-prints*, page arXiv:1110.3571, October 2011.

[Hes97] Lars Hesselholt. Witt vectors of non-commutative rings and topological cyclic homology. *Acta Math.*, 178(1):109–141, 1997.

[HHR16] M. A. Hill, M. J. Hopkins, and D. C. Ravenel. On the nonexistence of elements of Kervaire invariant one. *Ann. of Math. (2)*, 184(1):1–262, 2016.

[HL10] Michael Hill and Tyler Lawson. Automorphic forms and cohomology theories on Shimura curves of small discriminant. *Adv. Math.*, 225(2):1013–1045, 2010.

[HM03] Lars Hesselholt and Ib Madsen. On the K-theory of local fields. *Ann. of Math. (2)*, 158(1):1–113, 2003.

[HS98] Michael J. Hopkins and Jeffrey H. Smith. Nilpotence and stable homotopy theory. II. *Ann. of Math. (2)*, 148(1):1–49, 1998.

[HSW20] Jeremy Hahn, Andrew Senger, and Dylan Wilson. Odd primary analogs of Real orientations. *arXiv e-prints*, page arXiv:2009.12716, September 2020.

[HW20] Jeremy Hahn and Dylan Wilson. Redshift and multiplication for truncated Brown-Peterson spectra. *arXiv e-prints*, page arXiv:2012.00864, December 2020.

[Lau99] Gerd Laures. The topological q-expansion principle. *Topology*, 38(2):387–425, 1999.

[LH19] Thomas Nikolaus Lars Hesselholt. *Topological cyclic homology*. CRC Press, 2019.

[Lic73] Stephen Lichtenbaum. Values of zeta-functions, étale cohomology, and algebraic K-theory. pages 489–501. Lecture Notes in Math., Vol. 342, 1973.

[LN12] Tyler Lawson and Niko Naumann. Commutativity conditions for truncated Brown-Peterson spectra of height 2. *J. Topol.*, 5(1):137–168, 2012.

[MR99] Mark Mahowald and Charles Rezk. Brown-Comenetz duality and the Adams spectral sequence. *Amer. J. Math.*, 121(6):1153–1177, 1999.

[NS18] Thomas Nikolaus and Peter Scholze. On topological cyclic homology. *Acta Math.*, 221(2):203–409, 2018.

[Pst18] Piotr Pstragowski. Synthetic spectra and the cellular motivic category. 03 2018.

[Qui72] Daniel Quillen. On the cohomology and K-theory of the general linear groups over a finite field. *Ann. of Math. (2)*, 96:552–586, 1972.

[Rog00] John Rognes. Algebraic k-theory of

finitely presented ring spectra, 2000. Talk at Oberwolfach.

[Tho85] R. W. Thomason. Algebraic K-theory and étale cohomology. *Ann. Sci. École Norm. Sup. (4)*, 18(3):437–552, 1985.

[Wal84] Friedhelm Waldhausen. Algebraic K-theory of spaces, localization, and the chromatic filtration of stable homotopy. In *Algebraic topology, Aarhus 1982 (Aarhus, 1982)*, volume 1051 of *Lecture Notes in Math.*, pages 173–195. Springer, Berlin, 1984.

[Wil90] A. Wiles. The Iwasawa conjecture for totally real fields. *Ann. of Math. (2)*, 131(3):493–540, 1990.